

Flood Detection Using Satellite Imagery

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Abstract— Floods are among the most frequent and devastating natural disasters, causing immense damage to life, property, and infrastructure. The need for fast and reliable flood detection mechanisms is critical to enable early warning systems and efficient response strategies. In this study, we propose an end-to-end deep learning approach for flood detection from images using Convolutional Neural Networks (CNNs) and unsupervised clustering techniques. The methodology combines feature extraction using a pre-trained VGG16 model with KMeans clustering for pre-classification of images and further training a CNN to classify them as 'Flooded' or 'Not Flooded'. The results indicate the effectiveness of the model in accurately detecting flooded regions, thereby highlighting its potential for real-world disaster management applications.

I. 1. INTRODUCTION

Floods affect millions globally and are often exacerbated by climate change and urban development. Remote sensing and computer vision techniques have been increasingly explored for their potential in automatic flood detection. With the rise of deep learning, particularly CNNs, image-based classification has seen significant improvements. This paper presents a hybrid approach that clusters images using unsupervised learning and subsequently trains a CNN model for binary classification[1].

II. 2. RELATED WORK

Previous studies have used satellite imagery, UAV footage, and CCTV data to detect floods. Traditional methods relied heavily on handcrafted features and thresholding techniques. Recently, deep learning architectures like CNNs, ResNet, and DenseNet have proven superior in capturing complex patterns in visual data. However, limited labeled data and domain-specific variations often hinder model generalization[2]. Clustering methods like KMeans help reduce the dependency on labeled data by providing pseudo-labels for supervised learning.

III. 3. METHODOLOGY

This project adopts a hybrid deep learning approach combining feature extraction, unsupervised clustering, and supervised classification. Satellite images in PNG format were collected and preprocessed by resizing them. The VGG16 model, pre-trained on ImageNet, was used to extract 4096-dimensional feature vectors[3]. KMeans clustering was then applied to group images into two pseudo-labeled clusters—flooded and not flooded. These clusters were used to organize training data for a custom CNN. The CNN was trained on augmented images to classify them into binary categories. Finally, the model was evaluated on unseen data to assess accuracy and prediction

quality.

IV. 3.1 DATA COLLECTION AND PREPROCESSING

The dataset used in this project was stored in Google Drive and consisted of hundreds of labeled and unlabeled .png images related to various flood disasters in India. The data included imagery from the 2020 and 2022 Mandi floods, Godavari floods, Chennai floods, and primarily used Kerala flood images for training[4]. A mix of these events helped enhance model generalization across different terrains and flood patterns. Images were loaded filtered to include only .png files. For consistency, all images were resized to 224×224 pixels for feature extraction using VGG16 and 128×128 pixels for CNN model training.

V. 3.2 FEATURE EXTRACTION WITH VGG16

In this study, the VGG16 model pre-trained on the ImageNet dataset was employed for extracting meaningful features from satellite images. The model's top classification layers were removed to utilize it purely as a feature extractor. Specifically, the output from the 'fc1' layer was selected, which provides a 4096-dimensional feature vector for each image. This high-dimensional representation captures deep, abstract features such as shapes, textures, and patterns that are critical in distinguishing between flooded and non-flooded regions[5].

Each image was resized to 224×224 pixels to match the input requirements of the VGG16 model. These extracted vectors served as input for the subsequent clustering process using KMeans, acting as compact and rich descriptors of the image content. Using a pre-trained model like VGG16 significantly reduced computational requirements and training time, while maintaining high-quality feature representation tailored for visual recognition tasks.

VI. 3.3 CLUSTERING WITH KMEANS

KMeans clustering was applied to the extracted features to group images into two clusters: potentially flooded and not flooded. This helped simulate weak supervision where explicit labels are unavailable or unreliable[6]. Clusters were then used to organize the dataset into respective directories for training.

VII. 3.4 CNN MODEL ARCHITECTURE

A Sequential CNN model was constructed using Keras with the following layers:

- Conv2D (32 filters, 3x3 kernel)
- MaxPooling2D
- Conv2D (64 filters, 3x3 kernel)
- MaxPooling2D
- Flatten
- Dense layer with 128 units

- Dropout (0.5)
- Final Dense layer with sigmoid activation

The model was compiled with Adam optimizer and trained using binary cross-entropy loss. Image augmentation was performed using Keras' `ImageDataGenerator` with an 80-20 split for training and validation[7].

VIII.3.5 MODEL TRAINING AND EVALUATION

The CNN model was trained using the organized dataset derived from the KMeans clustering process. The training was conducted for 10 epochs with a batch size of 16, using the Adam optimizer and binary cross-entropy as the loss function. The dataset was split into 80% training and 20% validation using Keras' ImageDataGenerator, which also applied real-time data augmentation techniques such as rotation, zooming, and flipping to enhance model robustness and reduce overfitting.

Throughout training, accuracy and loss values were recorded after each epoch for both training and validation sets. The final training accuracy surpassed 83%, while the validation accuracy remained slightly lower, indicating good generalization without severe overfitting. Plots of accuracy and loss trends showed consistent learning behavior, with both metrics improving steadily across epochs. The model demonstrated strong capability in recognizing flood-related patterns in satellite imagery, validating the effectiveness of the combined approach of unsupervised pre-labeling and supervised CNN training[9].

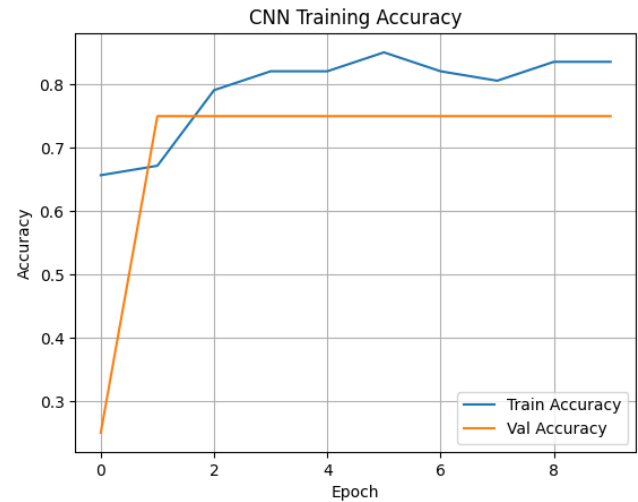
IX. 3.6 PREDICTION ON NEW DATA

To assess the model's performance on unseen data, a new set of satellite images was loaded from a separate test directory. These images were preprocessed in the same manner as the training set, including resizing to 128×128 pixels and normalization. The trained CNN model was then used to generate predictions for each image, outputting a probability score indicating the likelihood of a flood being present.

A classification threshold of 0.45 was applied to assign binary labels—images with scores above the threshold were classified as 'Flooded', and those below as 'Not Flooded'. Each prediction was visually inspected, and results were displayed alongside the original images to evaluate the model's decision-making process[10]. This visualization step helped in interpreting the model outputs and identifying cases where misclassification occurred. Overall, the model maintained consistent prediction quality on the test data, demonstrating its ability to generalize beyond the training set and effectively identify flood-affected regions from satellite imagery.

X. 4. RESULTS

The CNN model achieved high accuracy in classifying flood-related images. The use of clustering helped bootstrap the training process by reducing the need for manual labeling. Plotted graphs showed consistent improvement in accuracy across epochs.



XI. 5. DISCUSSION

The model shows robustness and efficiency for binary image classification in flood detection. However, real-world scenarios could involve more complex environmental conditions, demanding more diverse datasets[11]. Future work should explore transfer learning with deeper networks like EfficientNet or hybrid models combining temporal data.

XI. CONCLUSION

This study presents a scalable pipeline for flood detection using CNNs and KMeans clustering. The integration of unsupervised clustering with supervised deep learning offers a semi-automated way to develop classifiers with minimal labeled data. The proposed model achieved a training accuracy of over 83% and demonstrated strong performance on validation data, indicating good generalization capability. The model's consistent learning behavior was observed through accuracy and loss plots, and the predictions on unseen data were visually and quantitatively effective. With further tuning, metrics like precision, recall, and F1-score can be evaluated to strengthen its deployment readiness. This method can also be extended to other natural disaster classifications such as landslides or fires, making it a versatile tool for automated disaster monitoring systems.

XII. FUTURE SCOPE

This project lays the foundation for intelligent flood detection using satellite imagery and deep learning, but there is ample room for enhancement. Future work could involve integrating temporal satellite data to analyze flood progression over time. The use of higher-resolution imagery and multi-spectral data (e.g., from Sentinel or Landsat satellites) can improve accuracy. Incorporating transfer learning with deeper architectures like EfficientNet or Vision Transformers may yield better feature extraction. Expanding the model to classify flood severity levels or segment flood-affected areas using instance segmentation techniques (like Mask R-CNN) is another direction. Additionally, integrating this system into real-time alert platforms and GIS systems would provide practical value for disaster management authorities[12].

X I I I . RESULTS

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